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POSTER ABSTRACTS

722.ALLOGENEIC TRANSPLANTATION: ACUTE AND CHRONIC GVHD AND IMMUNE RECONSTITUTION

Loss of Microbiota-Derived Metabolites Is Associated with Mucosal-Barrier-Injury Bloodstream Infection in Patients Undergoing Allogeneic Hematopoietic Stem Cell Transplantation

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Background

Conditioning for allogeneic hematologic stem cell transplantation (allo-SCT) is associated with mucosal barrier injury (MBI) and neutropenia. This results in a high risk of infection and, consequently, in higher mortality, with more than 40% of allo-SCT patients suffering from bloodstream infections (BSI) in the first 30 days (Russo et al. 2018; Carreira et al. 2022). Allo-SCT is also often accompanied by disruption of the intestinal microbiome and microbiota-derived metabolites, negatively affecting clinical outcomes. (Thiele Orberg et al. 2024). These conditions facilitate bacterial translocation from the gut into the bloodstream and cause bloodstream infections (MBI-BSI). However, microbiome-related factors for translocation are still not fully understood. In patients undergoing allo-SCT, we assessed the impact of the intestinal microbiome and microbiota-derived metabolites on the development of MBI-BSI.

Methods

This prospective, longitudinal study was conducted at the Technical University of Munich, Germany, from 08/2019 to 03/2022 and included 85 patients receiving allo-SCT. Stool and saliva samples were collected weekly from the day of initial admission up to 4 weeks after allo-SCT, and blood cultures (BC) upon the occurrence of fever >38,5°C. Parallel to bio-sampling, we recorded infections, GvHD occurrence, and mortality and administered antibiotics. We analyzed microbiome profiles in all samples using 16S rRNA amplicon sequencing to assess genus-level taxonomic composition and monitor domination by pathobionts. Metagenomic sequencing (MGS) was used for strain-level taxonomic analysis in stool samples obtained around BSI and the corresponding blood culture sample. In order to assess potential translocation, we matched metagenome-assembled genomes in blood and stool using the average nucleotide identity score (ANI). Furthermore we evaluated changes in fecal concentrations of short-chain fatty acids (SCFAs) and branched-chain fatty acids (BCFAs) via targeted mass-spectrometry. We compared the BSI patients to allo-SCT patients without BSI ("control" subgroup) during different time points depending on when the BSI occurred. Metabolite concentration levels were measured 10 days before the BSI occurred, as well as 2 days during the BSI and 10 days after the BSI.

Results

Of 85 allo-SCT patients, 41 (48%) developed BSI confirmed via positive BC testing. Nine of those 41 patients (22%) could be excluded due to BSI stemming from sources other than MBI. Intestinal translocation was the likely source of BSI in the

remaining 32 cases (78%) ("BSI" subgroup). In 15 of 32 patients (37%), intestinal domination >30% of a single taxon [e.g., *Bacteroides*] was detected. In 5 of the 15 patients, the BSI-identified pathogen was dominant (above 30% abundance). We observed reduced α -diversity in BSI patients compared to controls. Loss of SCFAs levels were observed, especially BCFAs, which were reduced in the BSI subgroup before the BSI compared to controls. In 23 patients (72%) of the "BSI" subgroup, the BSI pathogen could be recultivated and sampled for MGS. The pathogen could not be recultivated in 9 patients (28%). Via MGS, an ANI match > 99% in blood and stool was detected in 6 out of 23 patients (26%), indicating that the intestinal strain translocated into the bloodstream ("ANI" subgroup). Of those 6 ANI patients, 2 showed domination of the responsible pathogen and an aerobic intestinal milieu. The other 4 ANI patients displayed an anaerobic community and no pathobiont domination.

Conclusions

We observed a loss of BCFAs before MBI-BSIs, suggesting they may enhance barrier integrity, which requires further research. Domination of the responsible pathogen was not necessary for translocation. However, detecting specific signatures, such as an overabundance of specific pathobionts (such as *Enterococcus* or *E. coli*) and reduced BCFAs concentrations, could be useful as a biomarker for MBI-BSI. In patients with risk factors for MBI, like high-dose chemotherapy (Elad et al. 2022), such a biomarker may provide a basis for initiating prophylactic narrow-spectrum antibiotic treatment.

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